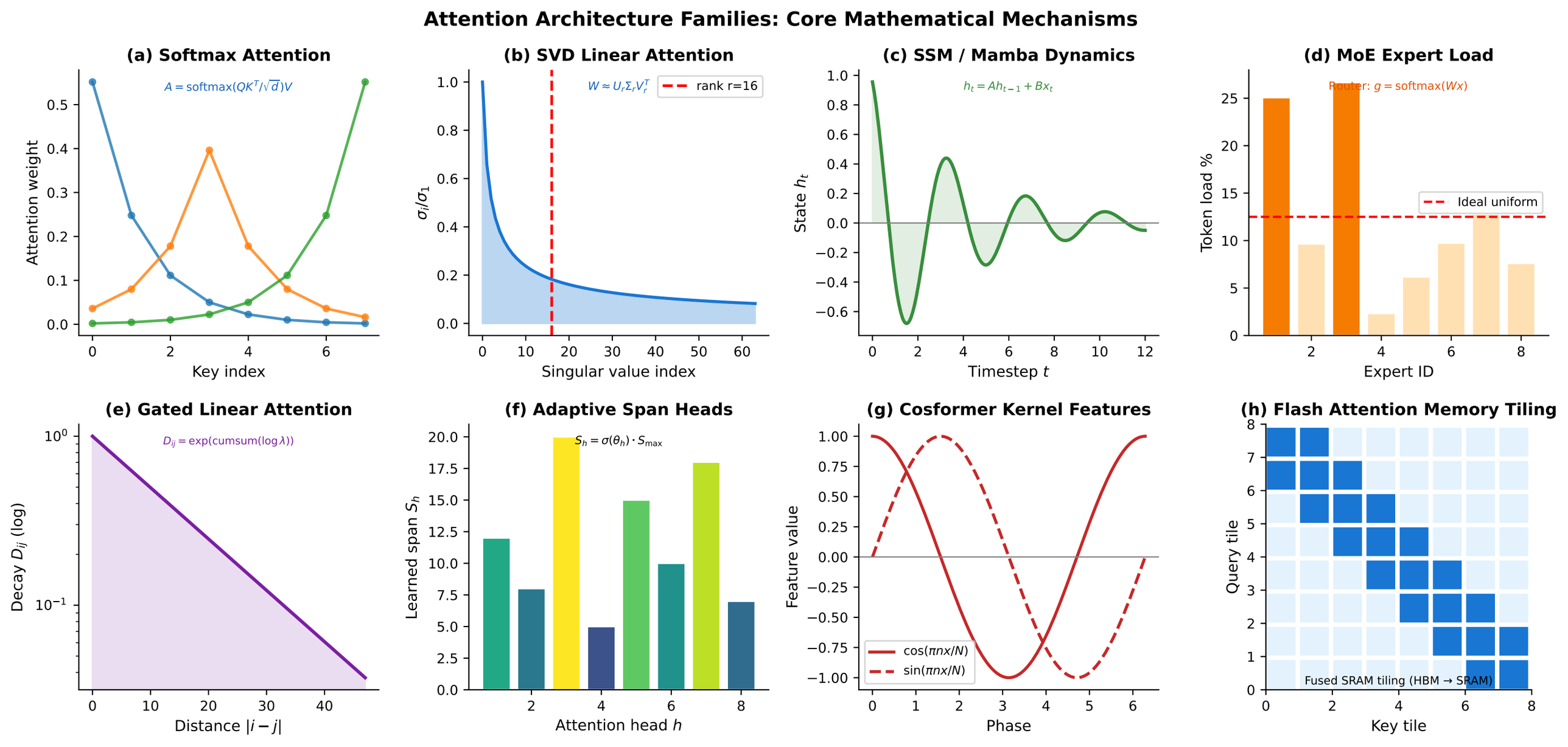

Attention Architecture Families: Core Mathematical Mechanisms
(a) Softmax Attention
$A = \text{softmax}(QK^T/\sqrt{d})V$
Attention weight
Key index
(b) SVD Linear Attention
$W \approx U_r \Sigma_r V_r^T$
rank r=16
σ_i/σ_1
Singular value index
(c) SSM / Mamba Dynamics
$h_t = Ah_{t-1} + Bx_t$
State h_t
Timestep t
(d) MoE Expert Load
Router: $g = \text{softmax}(Wx)$
Token load %
Ideal uniform
Expert ID
(e) Gated Linear Attention
$D_{ij} = \exp(\text{cumsum}(\log \lambda))$
Decay D_{ij} (log)
Distance $|i-j|$
(f) Adaptive Span Heads
$S_h = \sigma(\theta_h) \cdot S_{\max}$
Learned span S_h
Attention head h
(g) Cosformer Kernel Features
Feature value
$\cos(\pi nx/N)$
$\sin(\pi nx/N)$
Phase
(h) Flash Attention Memory Tiling
Query tile
Fused SRAM tiling (HBM → SRAM)
Key tile